

Prediction Steps

Step	What happens	Uses
1. Collect inputs	Every form field is gathered into a single-row table	pandas
2. Encode Yes/No fields	Gender, diabetic, and smoker are converted to the same codes used in training	Saved label encoders
3. Encode region	Region is expanded into the matching one-hot columns	Saved region encoder
4. Scale numbers	Age, BMI, blood pressure, and children are rescaled the same way as training data	Saved scaler
5. Match column order	Columns are reordered to exactly match what the model expects	Saved metadata
6. Predict	The trained model produces a single dollar estimate	Saved model
7. Build a range	The model's own historical average error is added and subtracted from the estimate	Saved test MAE